

SPECT Visual Interpretation (VI) Assessment

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Summary

Beginning in December 2024, Screening SPECT imaging acquired at PPMI imaging centers according to the PPMI imaging protocol was transferred to the XingImaging core lab for visual interpretation. SPECT imaging acquired prior to December 2024 was visually interpreted by Invicro, the previous PPMI imaging core lab. SPECT images visually assessed for dopamine transporter deficit were reviewed following the procedure outlined in the method section.

Method

The DAT SPECT scans for participants newly enrolled in PPMI Clinical in the Healthy Control and PD cohorts acquired at the Screening Visit using either 123I-DaTscan™ or 99mTc-TRODAT-1 were visually read by two expert, independent reviewers. If both readers agree, their interpretation determines the final classification for this scan. A third reader independently read the images in cases where there was discord between the two reader interpretations. In such cases, the final interpretation of the scan was based on the consensus of any two out of the three readers.

Scans were read as either showing evidence of dopamine transporter deficit (i.e. abnormal) or not showing evidence of dopamine transporter deficit (i.e. normal.). 123I-DaTscan™ was used as the primary radiotracer for DAT SPECT imaging. Select sites were approved for use of 99mTc-TRODAT-1 due to limited availability of 123I-DaTscan™.

Normal images are characterized by two relatively symmetric comma or crescent-shaped focal regions of radiotracer uptake mirrored about the midline in transaxial images. Striatal uptake, comprising both the caudate and putamen, is distinctly evident compared to surrounding brain tissue.

Abnormal images typically fall into at least one of the following three general categories:

- Activity is asymmetric, e.g. uptake in the region of the putamen of one hemisphere is absent or greatly reduced with respect to the other. Uptake is still visible in the caudate nuclei of both hemispheres resulting in a comma or crescent shape in one and a circular or oval focus in the other. There may be reduced uptake between at least one striatum and surrounding tissues.

- Ioflupane uptake is absent in the putamen of both hemispheres and confined to the caudate nuclei. The signal is relatively symmetric and forms two roughly circular or oval foci. Uptake of one or both is generally reduced.
- Uptake is absent in the putamen of both hemispheres and greatly reduced in one or both caudate nuclei. Uptake of the striata with respect to the background is reduced (1).

References

1. DaTSCAN™ [package insert]. Arlington Heights, IL: GE Healthcare; 2011.

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